STAT 330-001

40142034 32

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Entry: 12:11pm 10/21/2026 Checked out:

Employee: Joe R.

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly.
Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

NS

40142034

032

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

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Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

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(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

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(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

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(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

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(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

Because the coefficients are being shrunk. The model shrinks variable coefficients to reduce variance in exchange for a little bias on a controlled level.

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$\left(\frac{2.4}{4}\right)\left(\frac{0.25}{1-0.25}\right)$

$(0.6)(0.33)$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

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(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = \hat{\beta}_0 - \hat{Y} + \hat{\beta}_1 + \ldots + \hat{\beta}$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

β is the unknown coefficient for a variable.

$\hat{\beta}$ is the predicted coefficient for a variable.

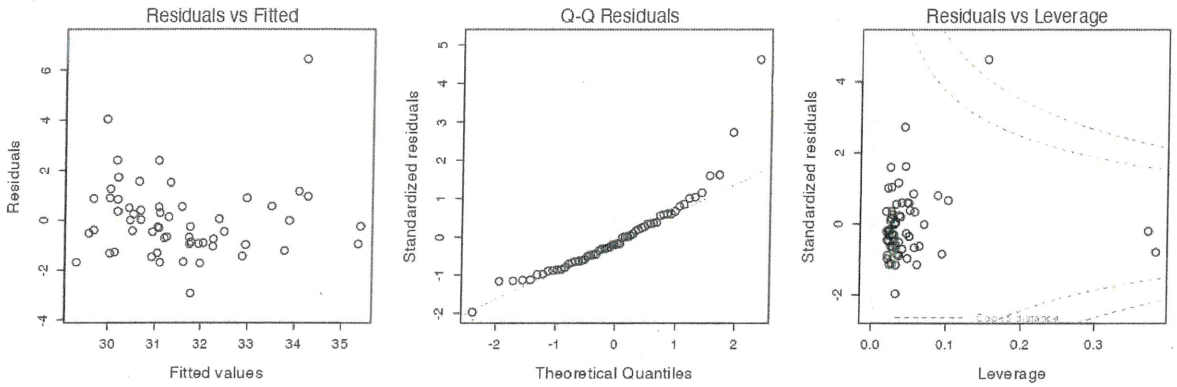
(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$



Coefficient	Estimate	Std. Error	t value	Pr($>\|t\|$)
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

$\beta_1 = 0.472892$

Hold all else constant, each inch increase in height increases the number of hotdogs eaten by 0.472892.

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

For height to hotdogs, yes, the data is approximately linear. For minutes to hotdogs, the data is more funnel shaped with some outliers that deserve more investigation/attention.

(B) There is independence between each observation

Yes, observations have no direct influence on each other.

(C) The errors are normally distributed

Yes, the errors are approximately normally distributed.

(D) The errors have constant variance (homoscedasticity)

The variance of the errors increases slightly on the high and low ends of the data set, showing slight homoscedasticity.

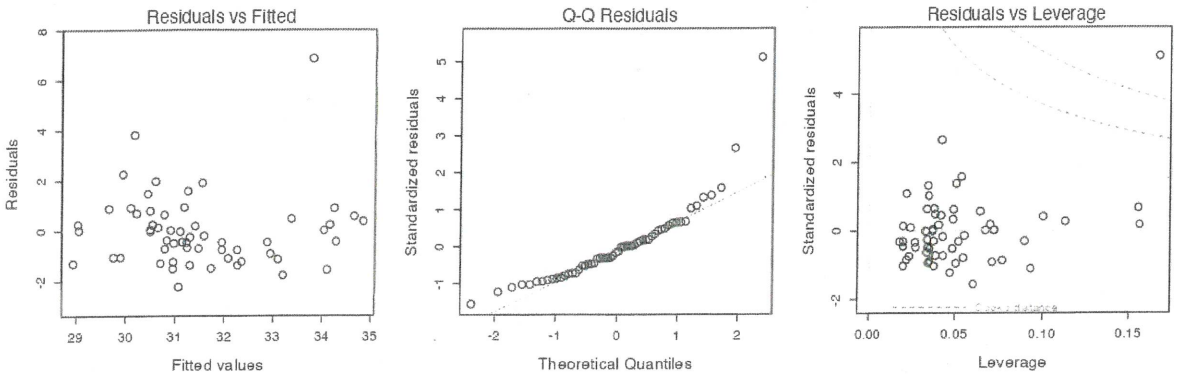
(E) There are no unduly influential points

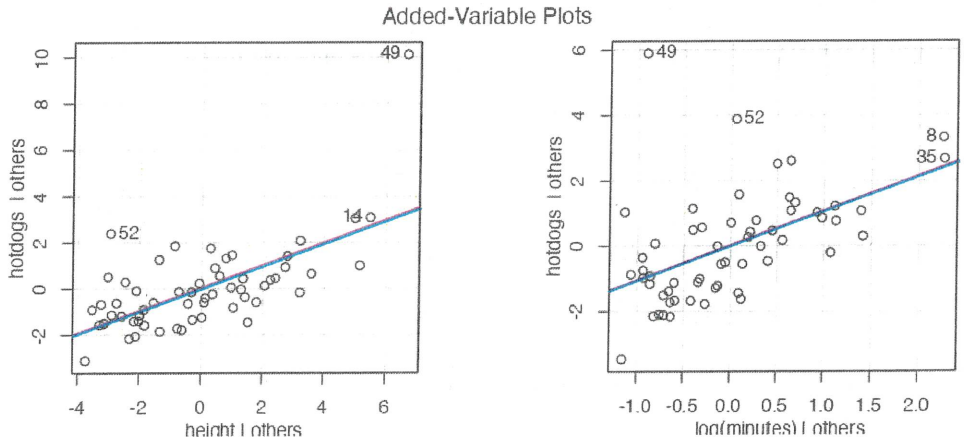
No, point number 49 may have unduly influence and requires investigation/attention.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

A lower value in a Breusch-Pagan test is typically better, so I would explain this to the researcher and ask how they reached their conclusion.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Model 2 provides Added variable plots that are better fitted to the data with better "Other summaries" scores. However, the Residuals vs Fitted and Q-Q residuals plots remain unchanged while the results have the added complexity of one variable being transformed.

Question 16: State the estimated value of β_2 and provide an interpretation in context

Holding all else constant, for every log(1) increase in minutes, the number of hotdogs eaten goes up 1.06702.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\hat{y} = \hat{\beta}_0 + \beta_1 \text{height} + \beta_2 \text{minutes}$

$\hat{y} = -6.05989 + 0.48834(68) + 1.06702(4.369)$

$\hat{y} = 31.81$ hotdogs

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

30.9 – 33.15

Part III - Application

Questions 20-24 concern the following dataset:

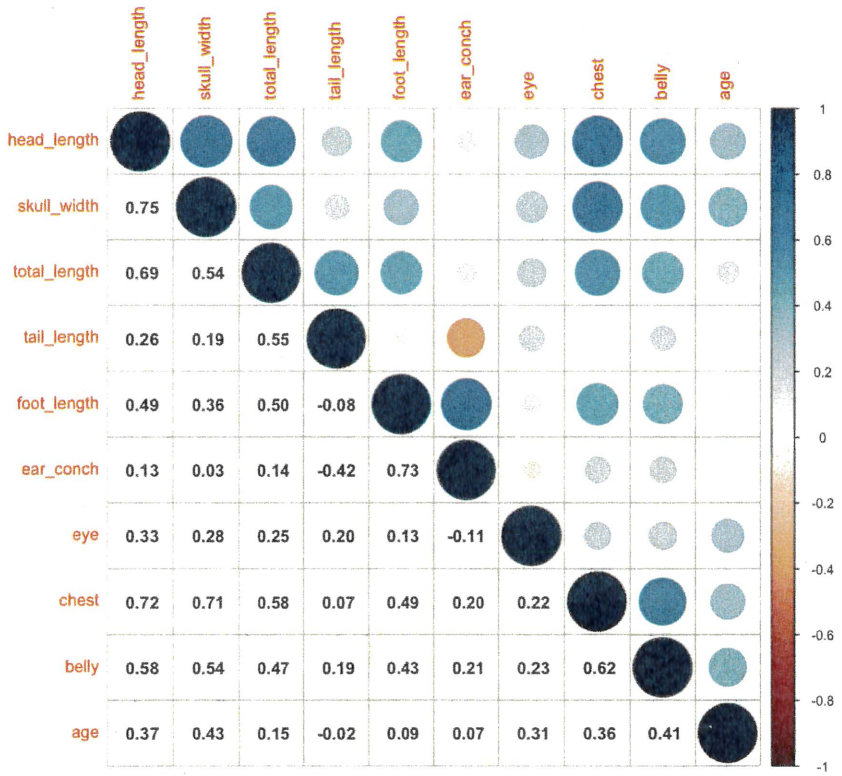
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
foot_length   -0.21499    0.10733  -2.003   0.0507 .
ear_conch      0.15730    0.10231   1.538   0.1306
eye            0.43736    0.22549   1.940   0.0582 .
chest          0.01883    0.18154   0.104   0.9178
belly          0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$2.01 \pm 49 * 0.16559$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

The predictors individually are not significant, but their combination and influence on each other is.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

It selects the predictors that explain the most while getting rid of others, trading complexity for significant variables only to produce simpler equations.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

A Average RMSE across all test-training splits and report

D Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

C Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

D Initialize vectors for each model that you will use to store their RMSEs

C Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

C Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

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Hooton, Jonathan Paul

618151070 jh00t0n 623-806-9745

Test: 639623 Form: A Wadsworth, Spencer Gordon

Entry: 11:12am 10/21/2026 Checked out:

Employee: Mikayla T.

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The residuals are being shrunk towards 0

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(B) What is the conceptual difference between $\widehat{\beta}$ and β?

β is a true population mean, $\hat{\beta}$ is our estimate of that mean based on our sample data

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

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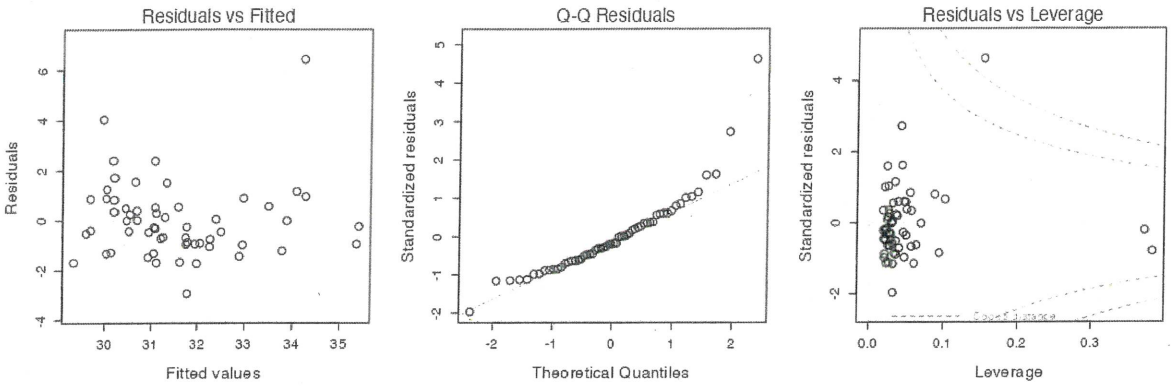

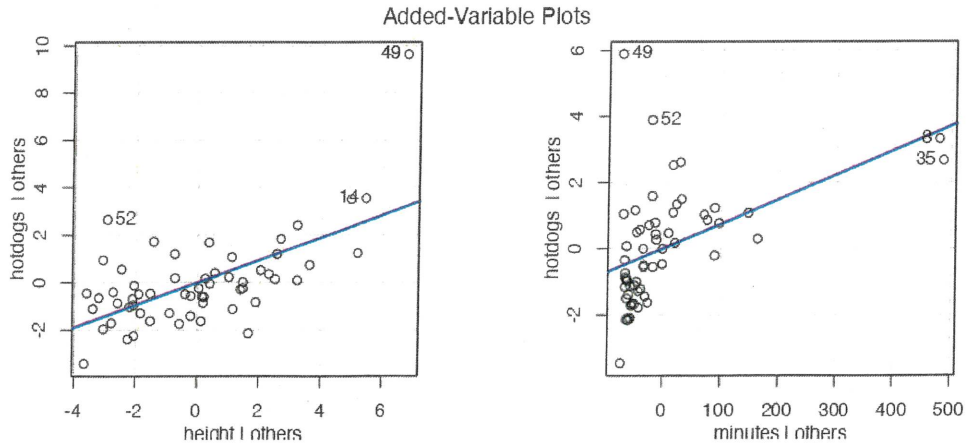

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Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

(.007417) ± (3.963 x 0.001871)

0.007414177

(0.000002, 0.014833)

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

$B_1 \approx .4729$

Holding B_2 constant, each additional 1 unit (inch) of height leads to an increase of 0.4729 hot dogs eaten

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

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Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

The residuals vs fitted plot is somewhat linear, the added variable plot for height looks quite linear, but the added variable plot for minutes doesn't really look linear.

(B) There is independence between each observation

I don't think we can assume independence, there's no randomness in the experimental design and most subjects are from the same counties

(C) The errors are normally distributed

The very low Shapiro-Wilk P value shows that this data is probably not normally distributed

(D) The errors have constant variance (homoscedasticity)

The Breush-Pagan value is high enough and the residual plot lacks the distinctive funnel shape, so this condition is probably satisfied

(E) There are no unduly influential points

There are definitely points that have more than their fair share of leverage and are past cook's distance

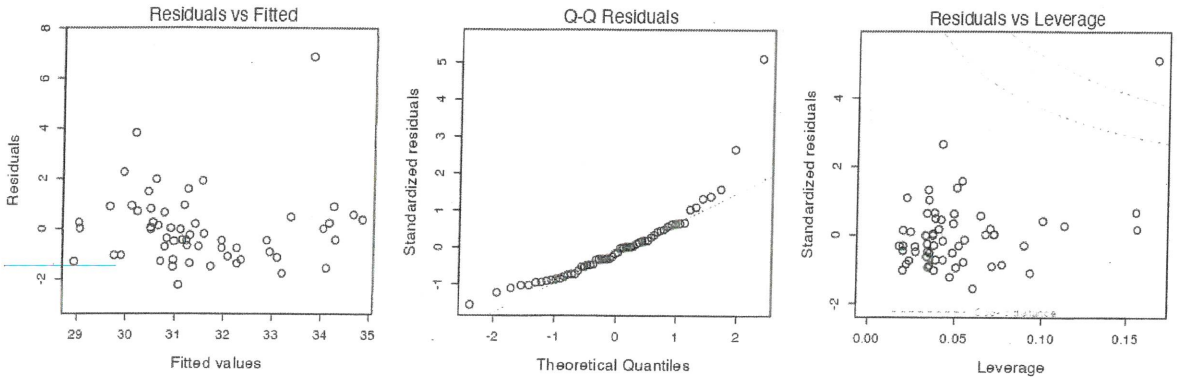
$\frac{2p}{n} = \frac{6}{54} \approx .103$

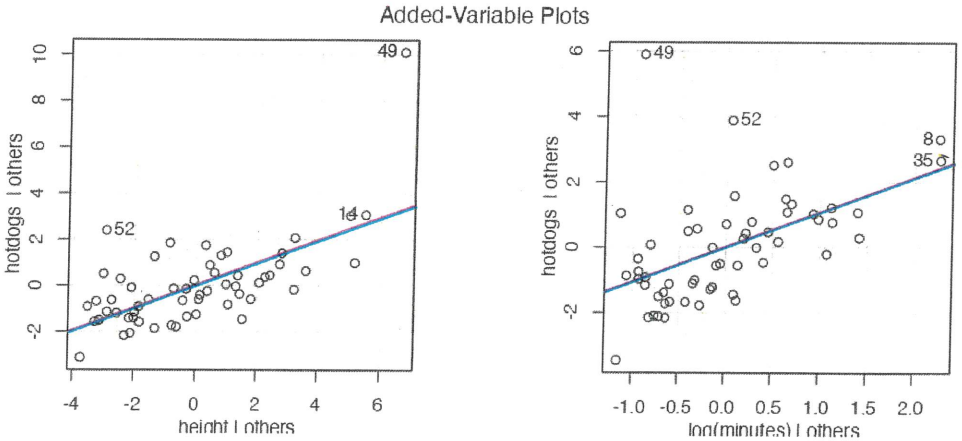
$\frac{4}{n} = \frac{4}{58} \approx 0.069$

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

so we can't assume equal variance

Model 2 seems to violate homoscedacity while model 1 doesn't, but we should still check other factors before saying which model is better

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

- the added variable plot for 2 looks much more linear
- individual points in model 2 don't have nearly as much leverage
- CV RMSE is slightly lower, but not much
- Breusch-Pagan value is lower, so we can no longer assume equal variance

Question 16: State the estimated value of β_2 and provide an interpretation in context

$B_2 = \log(\text{minutes}) \approx 1.06702$

$e^{1.06702} \approx 2.9067$?

$\log(1.06702) \approx 0.06487$?

Holding B_1 (height) constant, every 1 additional minute of practice leads to an increase of about 0.06487 hot dogs eaten

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that $\log(79) = 4.369$. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

Part III - Application

Questions 20-24 concern the following dataset:

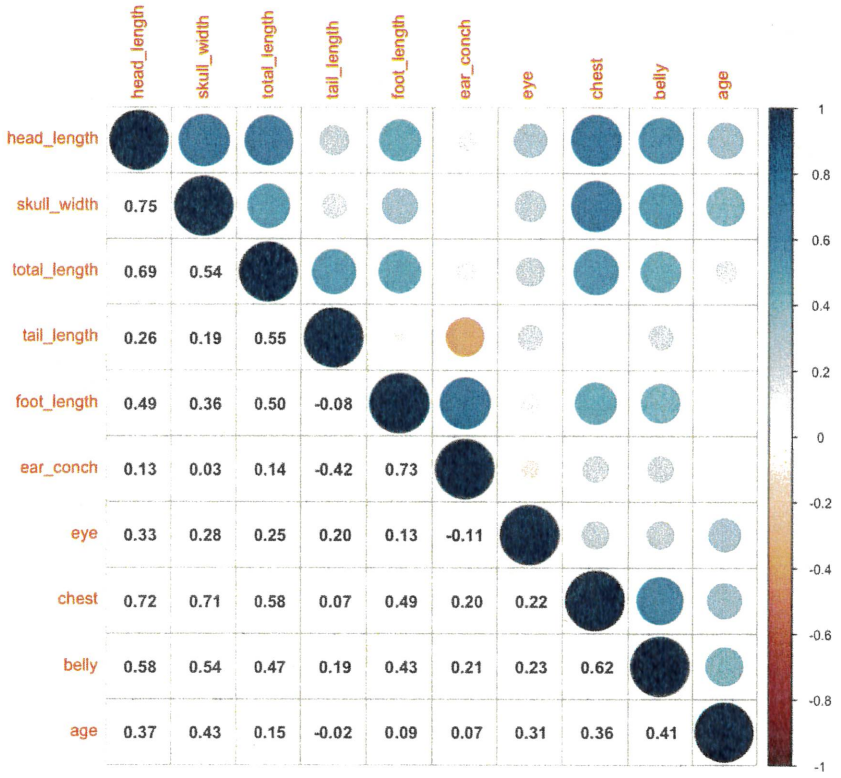
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
foot_length   -0.21499    0.10733  -2.003   0.0507 .
ear_conch      0.15730    0.10231   1.538   0.1306
eye            0.43736    0.22549   1.940   0.0582 .
chest          0.01883    0.18154   0.104   0.9178
belly          0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

.16559
.10549

$.16549 \pm (2.01 \cdot .10549)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the true relationship in cm/year of australian possums is within (-0.023, 0.443). Because this interval includes 0, we cannot prove any correlation between these variables

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

Individual coefficients have too much variance to have a significant correlation, but when you put them together they do

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

Best subsets finds the predictive value of every combination of predictors, then gives a mild penalty (2K) based on the number of predictors to prevent choosing a model that overfits

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

C Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

B Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

B Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

40142034 30

Scott, Marla

191490498 mkscott5 571-509-8701

Test: 639540 Form: A Wadsworth, Spencer Gordon

Entry: 10:38am 10/21/2026 Checked out:

Employee: Annabelle H.

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly.
Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

NS

40142034

030

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

The variance is being shrunk through the trade of small bias. The predictors are standardized prior to running Ridge/ Lasso.

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$\frac{(2.4)^2}{3+1}\left(\frac{0.25}{1-0.25}\right) = \frac{5.76}{4}(0.33)$

$1.44(0.33) = 0.4752$ ← D_i

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = (X^TX)^{-1}X^TY$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

β is the population parameter (unknown).

$\hat{\beta}$ is the estimated population parameter

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

Least squares est: $\hat{\beta} = (X^TX)^{-1}X^TY$

Least squares: $\beta = (Y - X\beta)^T(Y - X\beta)$

↓ take derivative with respect to β

$\left(\frac{\partial SSE}{\partial \beta} = -2X^TY + 2X^TX\beta = 0\right)$

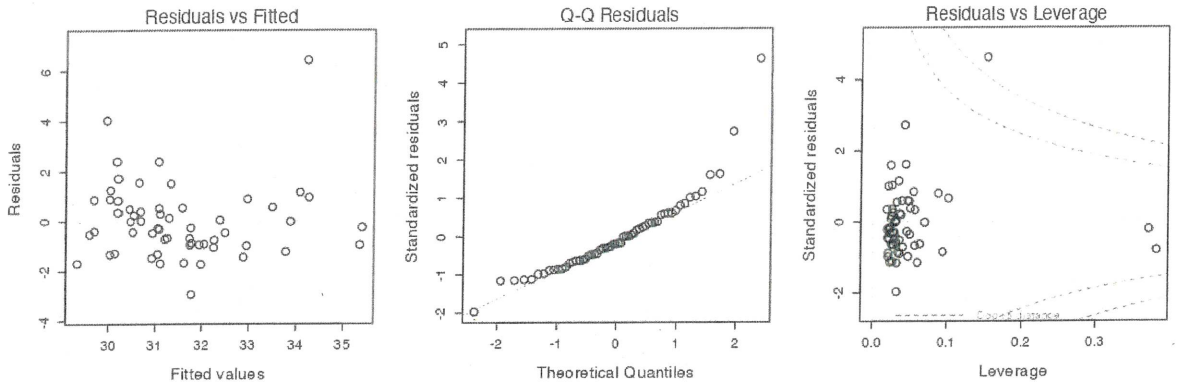
Taking the derivative of the least squares with respect to β yields the gradient.

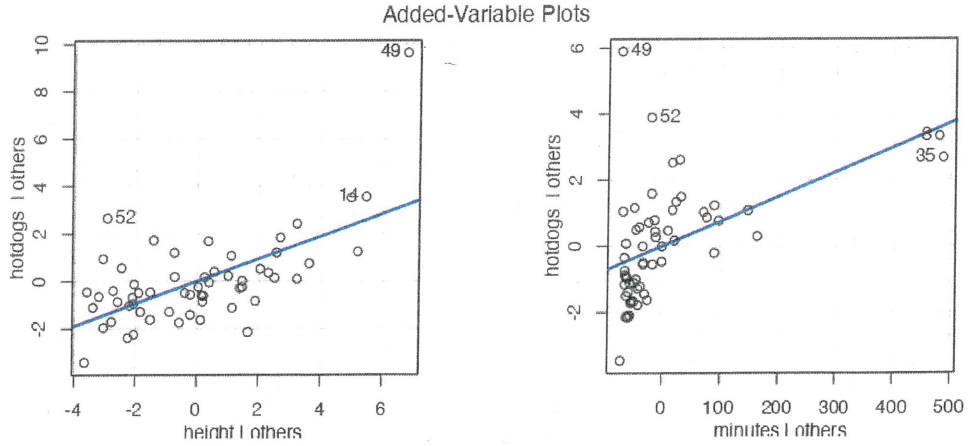
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	Pr($>\|t\|$)
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

$\hat{\beta}_1$ $\hat{\beta}_1 = 0.472892$

Holding all other predictors constant, each additional increase in height (inch) is associated with an estimated 0.47 increase in average hot dogs eaten.

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

height + mins hot dogs

Checking the residuals vs. fitted plot, there is a band around 0, but it doesn't seem completely patternless.

(B) There is independence between each observation

$Cov(\epsilon_i, \epsilon_j) = 0$

Based on the context of the problem, each observation is somewhat independent b/c they're all different people, but most of them are from the same 4 countries.

(C) The errors are normally distributed

$\epsilon_i \sim N(0, \sigma^2)$

Checking the qq plot, the values seem to follow the reference line for the most part, besides the edges. Also looking at Shapiro Wilk test

(D) The errors have constant variance (homoscedasticity)

Checking the residuals vs. fitted plot, but this time the vertical spread, which doesn't show constant variance. Also looking at Breusch-pagan test (p value > 0.05)

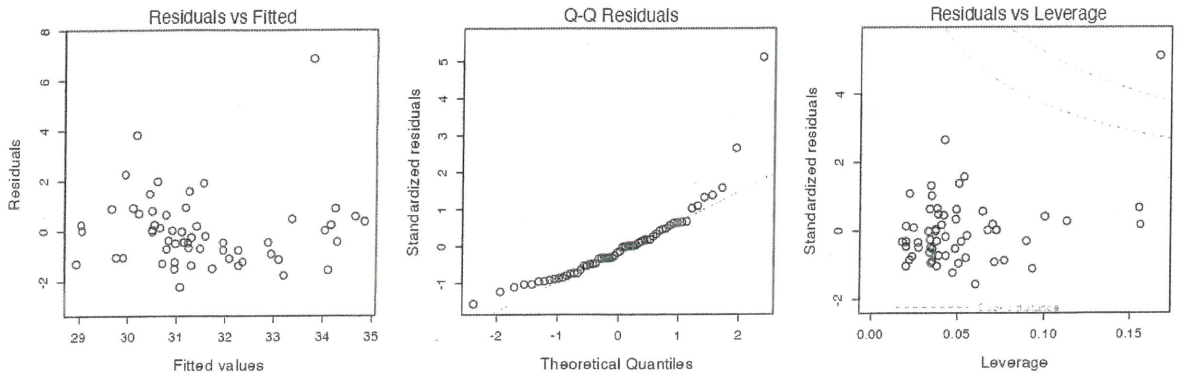
(E) There are no unduly influential points

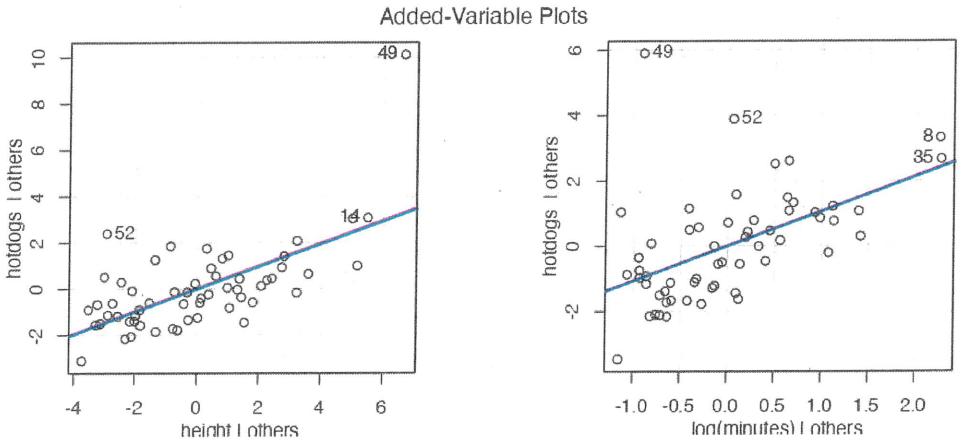
There are influential points as observed in the Residuals vs. leverage plot. There is a point in the top right corner on the other side of the dotted lines

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
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Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

(p value for Breusch-pagan)

I would say that one result ^ between two models doesn't automatically guarantee one model is superior. They need to analyze other pieces before coming to such a strong conclusion.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Model 2 provides a better Residuals vs. leverage visual as the points are less grouped, and on the left. Additionally, now Breusch-pagan p-value < 0.05. A disadvantage is that the intercept p-value increased in model 2.

Question 16: State the estimated value of β_2 and provide an interpretation in context

$\hat{\beta}_2 = 0.48834$

Holding all other predictors constant, each increase in an inch (height) is associated with an estimated 0.48834 increase in average hot dogs eaten.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$$\text{hotdogs} = \beta_0 + \beta_1 \text{height}_i + \beta_2 \log(\text{mins}) + \epsilon_{ij}$$

$$= 1 + \beta_1(68) + \beta_2(4.369) + \epsilon_{ij}$$

$$= -6.05599 + 0.48834(68) + 1.06702(4.369)$$

$$= -6.05599 + 33.20712 + 4.6618$$

$$= 31.81$$

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

$n-p-1$

(A) 55

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

There is a 95% probability that on average, my friend could eat between 31.61 and 33.295 hotdogs in 10 minutes.

Part III - Application

Questions 20-24 concern the following dataset:

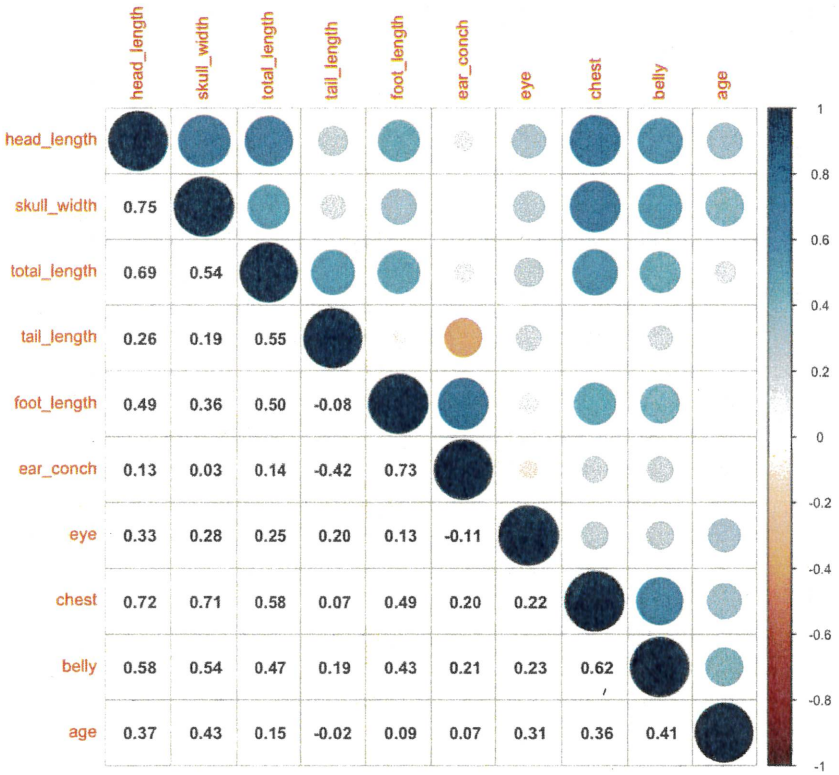
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

Estimate : 0.16559

$0.16559 \pm 0.95 * 0.975$

$0.16559 \pm 0.95 * 0.975$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

There is a 95% probability that on average, the belly width is between 0.18683 and 0.65283 cm.

Belly estimate: 0.20983

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

The multicollinearity between the individual model coefficients inflates the SE while keeping overall model looking fine.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

The best subsets algorithm using AIC as its criterion performs variable selection because AIC is focused more on predictors and provides a larger model. Best subsets finds global optimum and it's better to have a larger model when doing SD.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

C Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

C Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

40142034 29

Bemis, Alicia

265220249 aliciab4 8042998424

Test: 639428 Form: A
Entry: 09:47am 10/21/2026
Employee: Annabelle H.

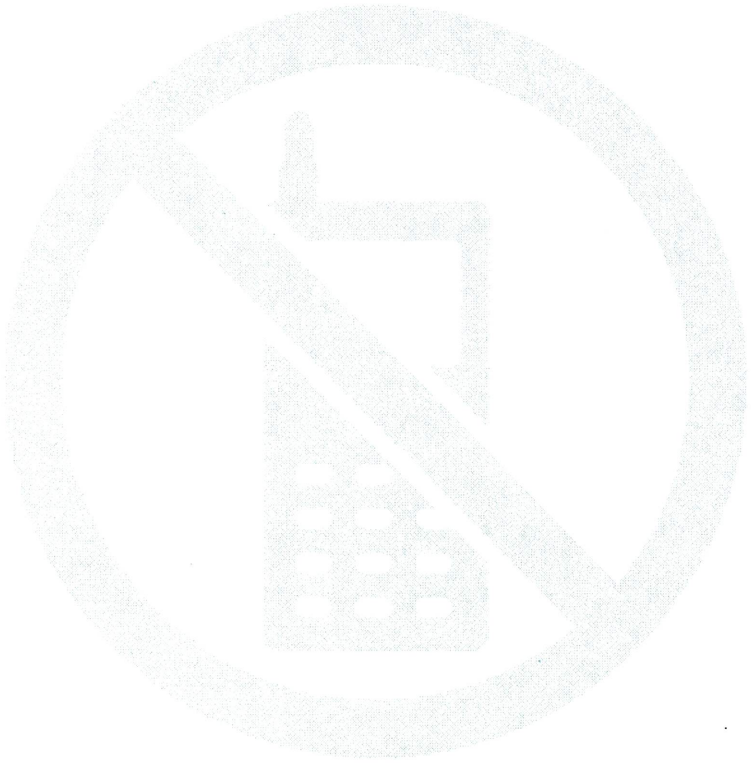
Wadsworth, Spencer Gordon
Checked out:

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE



Alicia Bemis

NS

40142034

029

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: **(Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: **(TRUE/FALSE)** If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: **(Circle one answer)** When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: **(Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality. → not independence

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression. equal variance assumption has to do w/ variance across predictors

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS they introduce bias, not reduce

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

LASSO & ridge regression are sometimes referred to as "shrinkage" methods because they both shrink the coefficient estimates. They accomplish this by introducing bias in two distinct ways into the model, which helps reduce the total error of the model. They use a lambda value which is determined by the lowest Cross-Validation RMSE (at least for the minimum lambda value) to help know how much bias to introduce into the model without losing the predictive performance of the model. This process ultimately results in the coefficient estimates shrinking.

$p+1=4$

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$= \frac{2.4^2}{4}\left(\frac{0.25}{0.75}\right) = \frac{5.76}{4}\left(\frac{0.25}{0.75}\right) = 1.44(1/3) = 0.48$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

~~(B)~~ $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

~~(C)~~ $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

~~(D)~~ ~~$D_i = 0.2$.~~ Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

~~(E)~~ ~~$D_i = 0.2$.~~ Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

~~(F)~~ ~~$D_i = 0.2$.~~ The observation is not influential because its leverage is less than 0.50.

~~(G)~~ Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$$\hat{\beta} = (X^TX)^{-1}X^TY$$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

β is the fixed coefficients, or the true population coefficients. It represents the parameter, while $\hat{\beta}$ represents the statistic. $\hat{\beta}$ is the fitted coefficients we have obtained by fitting our model on the sample data. Thus, β is the values of the coefficients which are fixed and from the population, whereas $\hat{\beta}$ is the values of the coefficients that are from our sample data.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

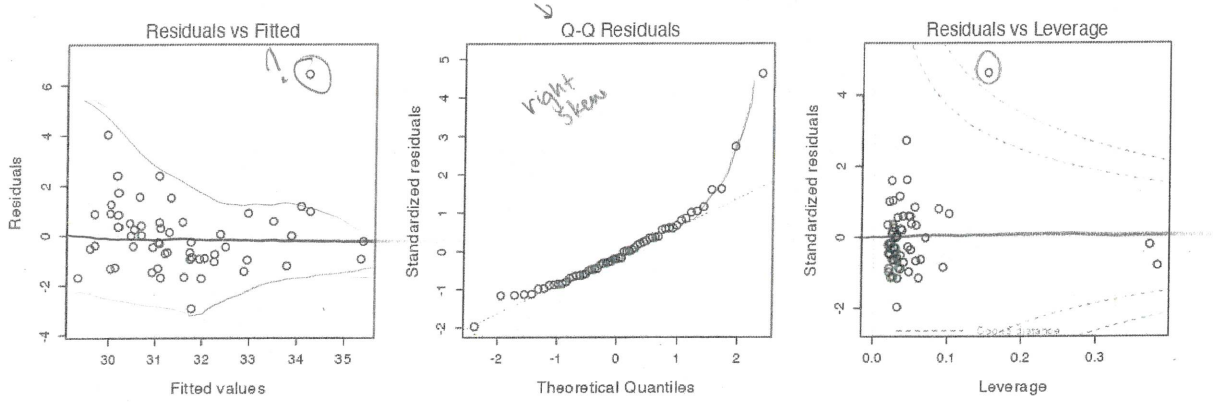
Least squares can be used to estimate regression coefficients. The matrix equation for the sum of squared errors is $SSE(\beta) = (Y-X\beta)^T(Y-X\beta)$. This is essentially creating a square matrix of the errors since $Y = X\beta + \varepsilon$, so $\varepsilon = Y - X\beta$. Then, we can expand out the $SSE(\beta)$ equation to get $Y^TY - 2\beta^TX^TY + \beta^TX^TX\beta$. Then, we want to find the smallest value of this expression, so we take the derivative of the expression and set it to zero. This is so that we can find the minimum of the expression, which will give us the smallest sum of square errors, which is ultimately our goal that allows us to find regression coefficients that minimize sum of square errors. After differentiating and setting to zero we get the normal equations: $X^TX\hat{\beta} = X^TY$. After multiplying both sides by $(X^TX)^{-1}$, we isolate $\hat{\beta}$ and get that $\hat{\beta} = (X^TX)^{-1}X^TY$. Ultimately, this process demonstrates that least squares is intrinsicly related to regression coefficients. Finding the smallest sum of square errors means that we are finding the [redacted] regression coefficients that will minimize this value.

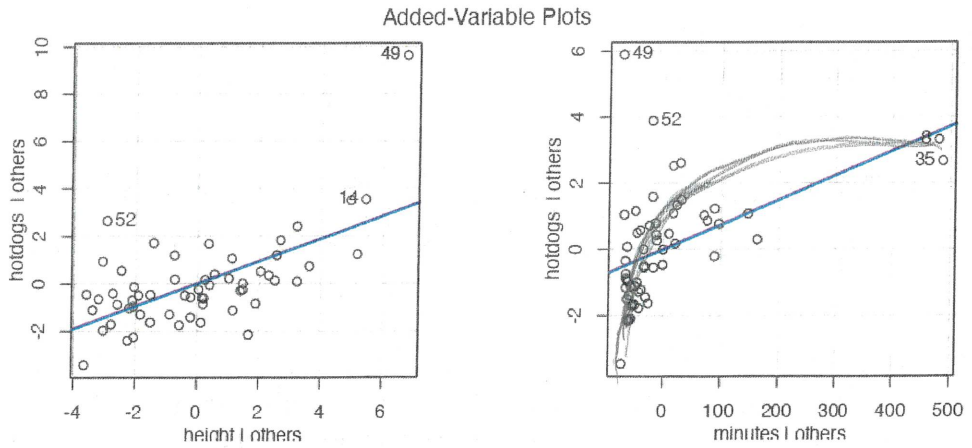
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

The estimated value of β_1 is 0.472892.

Interpretating: Holding minutes training per week constant, a one-inch increase in height is associated with an average increase in the number of hotdogs consumed in a 10-minute sitting of 0.472892 hotdogs.

Question 12: **(Circle all that apply)** Which of the following are true?

$CE = \beta_2 \pm 1.995(SE)$
$= 0.007417 \pm 1.995(0.001871)$
$= 0.007417 \pm 0.0037326$
$(0.0036844, 0.0111496)$

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Looking at the residuals vs. Fitted plot, we see that on the left side of the graph, there is a cluster of positive residuals, which we don't love because we want this plot to be pattern less about the line where residuals = 0. We also see that while the added variable plot for height looks good and pattern less, the plot for minutes has a clear and distinct pattern of curvature. Given this evidence, the data does not really satisfy the linearity assumption, so we would not want to move on with our model before checking to see if there is something we could do about that.

(B) There is independence between each observation

Our data was collected from hot dog eaters who were all competing at the same state fair eating competition. Thus, we would probably say that the independence assumption isn't met well by our data because the hot dog eaters could have influenced each other to eat faster or slower because they were in direct competition with each other.

(C) The errors are normally distributed

Looking at our Q-Q Plot, we see that our standardized residual points are deviating from the theoretical quantile line in an upwards direction on both the left and right side of the plot. This indicates that the distribution of our residuals is right skewed. Additionally, our Shapiro-Wilk test returned an incredibly small p-value, giving us more evidence that the distribution of our residuals is not normal. Therefore, we can reasonably say that our data has not met the normality assumption.

(D) The errors have constant variance (homoscedasticity)

First looking at the residuals vs. fitted plot again, we see that there is a small fanning / funneling shape such that there is greater spread in the data on the left of the plot than on the right. However, the p-value for our Breusch-Pagan test is 0.071, which is not too small. Given that the funneling shape is not extremely prominent, as well as the result of our Breusch-Pagan test, we can cautiously say that the data somewhat satisfies the constant variance assumption.

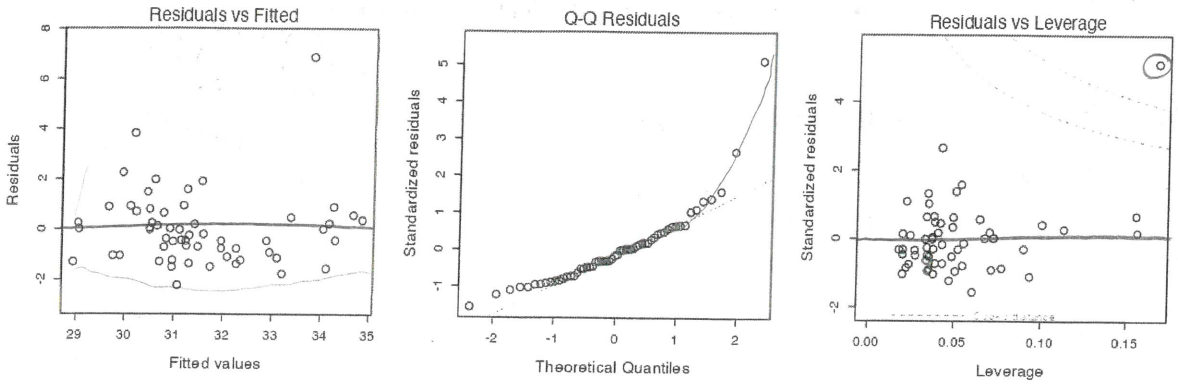
(E) There are no unduly influential points

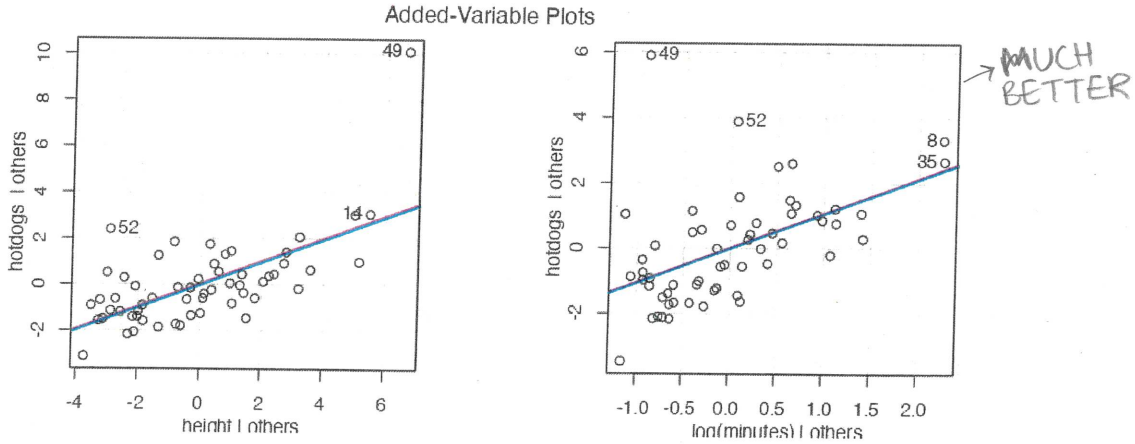
On the residuals versus leverage plot, we notice a point in the middle-top portion of the plot. This point is beyond the dashed lines of the Cook's Distance rule of thumb, meaning that this point has a very large Cook's Distance value. We would need to double check that this observation's data was input correctly, but if it is correct, we would likely say that model 1 does not meet the assumption that there are no unduly influential points in our regression. Note that this does not mean we would automatically delete this observation.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	Pr($>\|t\|$)
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value	
Breusch-Pagan p-value	0.028	→ smaller
Shapiro-Wilk p-value	3.43e-5	→ larger
CV RMSE	1.66702	→ smaller

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would note that the researcher is being short-sighted in two ways. They are first ignoring all other metrics we use to assess the several other assumptions we make with multiple linear regression. Thus, basing our decision about which model is better solely on one assumption is rash and short-sighted. The other reason this claim is shortsighted is because it only acknowledges one tool we use to assess the constant variance assumption. It does not take into account either a scale-location or residuals vs. fitted plot, instead observing only one metric to base a big decision on. Another reason why this is a poor claim is because it fails to acknowledge the nuance of using the p-values of hypothesis tests to assess our assumptions. It does not notice that the decrease in p-value from Model 1 to Model 2 is relatively small and that the p-value for Model 2 may not even be significant. Again, we would need to consider other tools to make a conclusion on just this one assumption for which model is better.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Advantages: The biggest advantage that Model 2 provides over model 1 is that the linearity assumption is met much better. We can see this most prominently in the difference between the added-variable plot for minutes from Model 1 to Model 2. Model 2 effectively eliminated the curvature that we saw for this plot with Model 1. Additionally, Model 2 has a smaller Cross-Validation Root Mean Squared Error. We also notice that the residuals vs. fitted plot includes less of a funnel / fanning shape, so we might think that the equal variance assumption may be fit better in Model 2.

Disadvantages: Along that same vein, however, the Breusch-Pagan p-value dropped to be less than 0.05, so we don't love that this p-value is smaller. That doesn't necessarily mean that model 1 is better at satisfying the equal variance assumption however because we still take into account the residual vs. fitted plot. In Model 2, we still have a problem with the normality assumption, with the Q-Q plot still showing a strong right skew in the residual distribution and the Shapiro-Wilk test still having a very small p-value. Finally, in Model 2, we still see the same influential point which needs to be investigated.

Question 16: State the estimated value of β_2 and provide an interpretation in context

The estimated value of β_2 is 1.06702.

Holding height constant, a 2.9067% increase in minutes spent training each week is associated with a 0.01244 increase in the average number of hot dogs eaten in a 10-minute sitting.

$e^{1.06702} = 2.9067$

$\log\left(1 + \frac{2.9067}{100}\right) = \log(1 + 0.029067) = \log(1.029067) = 0.01244$

height = 68
minutes = 79
log(79) = 4.369

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

→ No error included

hotdogs = $-6.05589 + 0.48834(68) + 1.06702(4.369)$
$= -6.05589 + 33.20712 + 4.66618038$
$= 31.81304$

My calculator says log(79) = 1.89776!
We should specify the base. I used the Note from the question about what log(79) equals though.

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55
(B) 57
(C) 56
(D) 48

$n - p - 1$
$= 58 - 2 - 1 = 55$

Question 19: You're given that ~~$s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$~~, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

estimate $\pm t^* \cdot SE$

$SE = 1.485$, $t^* = 2.004$

Use value from Q17: 31.81304

$31.81304 \pm 2.004(1.485) = 31.81304 \pm 2.97594$

$= (28.8371, 34.78898)$

Part III - Application

n=59

Questions 20-24 concern the following dataset:

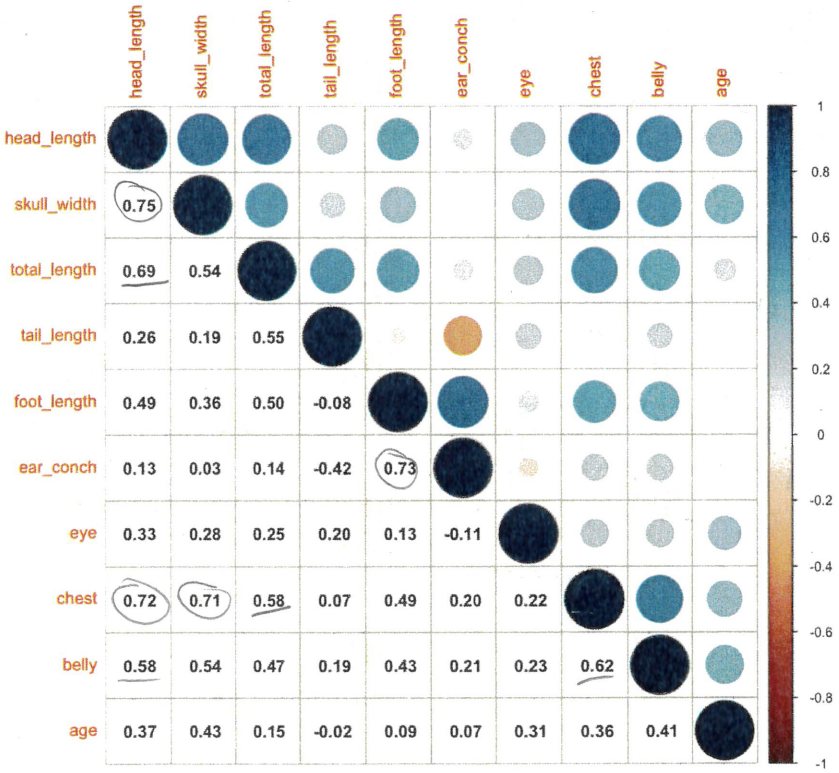
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$\beta_{s.w} = 0.16559$

$SE_{s.w} = 0.10599$

$0.16599 \pm (2.01)(0.10599)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

Holding all other predictors constant (head_length, skull_width, total_length, tail_length, foot_length, ear_conch, eye, and chest), we are 95% confident that each one-centimeter increase in belly girth is associated with a change in average age in years between a decrease of 0.023 and an increase of 0.443.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

I think the reason that we see this phenomenon is because we have strong evidence that at least one of our β regression coefficients is not equal to zero and thus we have evidence that at least one predictor is linearly associated with our response age, this does not mean that there should be a significant p-value for any of the individual model coefficients. The significant F-test indicates that the model at least has some value for us, but it could be that the combination of certain coefficients for predictors makes the F-test significant. With our individual coefficient p-values, no other predictors are considered, so we can justifiably see this phenomenon because the two tests aren't measuring the same thing

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

The "best subsets" algorithm using AIC goes through every model that is possible given our p predictors and calculates the AIC for each model. That means that this algorithm searches through 2^p different models, calculating the AIC for each one. It then chooses the best model by comparing the AIC values and finding the model with the best (lowest) AIC value.

$AIC = n\log(SSE/n) + 2k$

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

AIC & Adjusted-R^2 penalize predictors differently, so they are allowed to pick different models as best

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report ✓

D Average the predictions and record their value – No need for this

D Choose the model with the largest RMSE – We don't want this model

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

40142034 28

Gee, Emma Jane

529904467 emmagee2 801-661-0845

Test: 639352 Form: A Wadsworth, Spencer Gordon

Entry: 08:53am 10/21/2026 Checked out:

Employee: Annabelle H.

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

40142034

NS

028

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: **(Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

unique info not explained by the other variables of your X.

Question 2: **(TRUE/FALSE)** If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: **(Circle one answer)** When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: **(Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

Because they shrink the estimated coefficients values ($\hat{\beta}$'s) closer to 0 this is accomplished by introducing a little bias, by adding to what we're trying to minimize. WITHOUT SHRINKAGE WE'RE JUST TRYING TO MINIMIZE THE SUM of squared residuals, but when we use a shrinkage method we add a term that penalizes larger coefficients so the coefficients have to be lower to minimize it. How big our penalty is (λ) is chosen by cross-validation.

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$\frac{2.4^2}{4}\left(\frac{.25}{1-.25}\right) = 1.44(.3333) = .48$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

β IS THE UNKNOWN TRUE COEFFICIENT PARAMETER FROM THE POPULATION.

$\hat{\beta}$ IS THE ESTIMATE OF β, CALCULATED FROM A SAMPLE.

β IS FIXED BUT $\hat{\beta}$ CAN VARY DEPENDING ON THE SPECIFIC SAMPLE.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

WE WANT TO MINIMIZE THE SUM OF SQUARED ERRORS, WHICH MEANS THE LINE WILL BE AS CLOSE TO THE POINTS AS IT CAN GET. TO DO THIS WE TAKE THE $SSE(\beta) = (y - X\beta)^T (y - X\beta)$ AND EXPAND IT. THEN TO MINIMIZE SOMETHING, YOU TAKE THE DERIVATIVE AND SET IT EQUAL TO $\emptyset$.

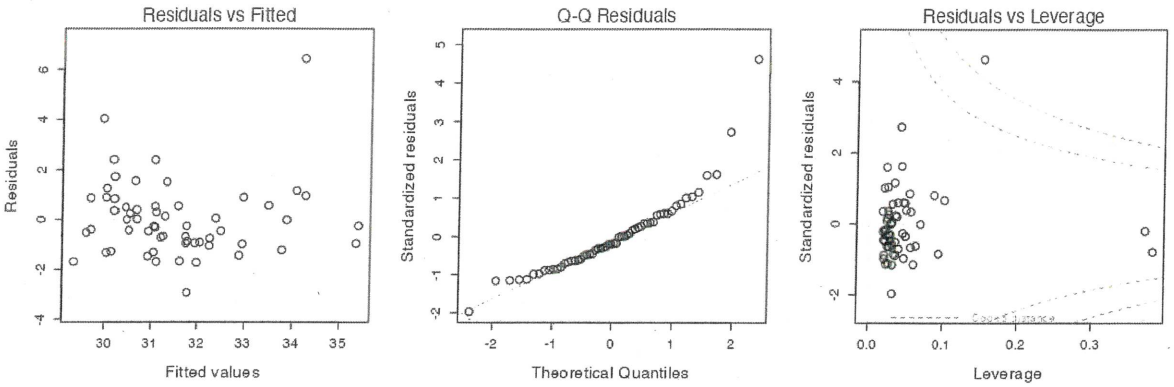
SO WE TAKE THE DERIVATIVE OF $SSE(\beta)$ WITH RESPECT TO β AND WE GET $-2X^T y + 2X^T X\beta = 0$ SO THEN TO SOLVE FOR THE β'S THAT WILL MINIMIZE IT, WE ADD THE FIRST TERM TO BOTH SIDES THEN DIVIDE BY 2 TO GET $X^T X\hat{\beta} = X^T y$. THEN BECAUSE A MATRIX MULTIPLIED BY IT'S INVERSE IS THE IDENTITY MATRIX, WE CAN MULTIPLY BOTH SIDES BY THE INVERSE OF $X^T X$ TO GET $(X^T X)^{-1} X^T X\hat{\beta} = (X^T X)^{-1} X^T y$ SO THEN WE CAN SOLVE FOR $\hat{\beta} = (X^T X)^{-1} X^T y$ TO GET THE VALUES OF β

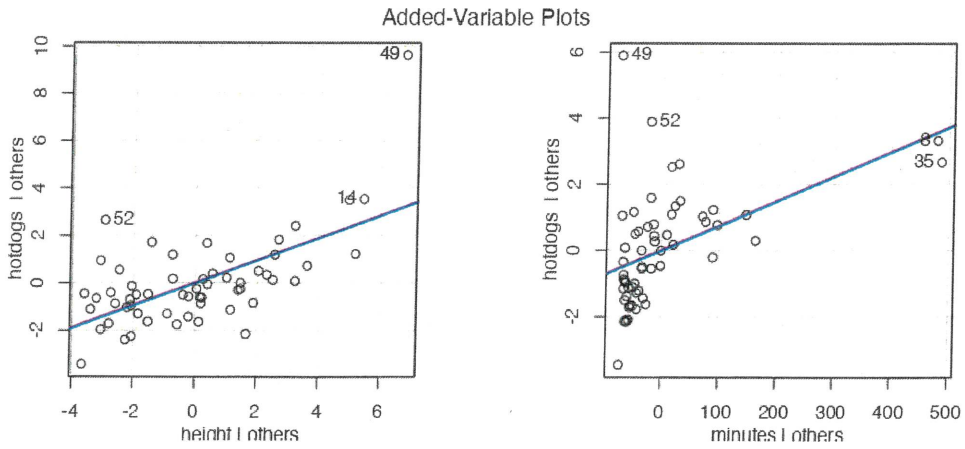
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

The estimated value of β_1, or $\hat{\beta}_1 = .472892$

THIS MEANS THAT HOLDING HOW MANY MINUTES A PERSON SPENT TRAINING PER WEEK CONSTANT, EACH ADDED INCH OF HEIGHT for a contestant IS ASSOCIATED WITH AN ESTIMATED .472892 INCREASE IN THE AVERAGE AMOUNT OF HOT DOGS CONSUMED IN A 10 MINUTE SITTING ON AVERAGE.

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

not very plausible because the residuals vs fitted values plot shows a curved relationship and looking at the added variable plots we see that height seems to have a linear relationship w/ hotdog consumption but minutes training has a curved relationship after accounting for height that isn't explained by the linear model.

(B) There is independence between each observation

not entirely satisfied: it might be not a big deal but since it wasn't random and several people came from each county, people from the same county might tend to be better or could be related and could affect consumption ability

(C) The errors are normally distributed

not really satisfied. The shapiro wilk test's pvalue is extremely small, meaning we have convincing evidence that the errors don't come from a normal distribution. and since the qq plot also shows a curved line, with points that deviate on both ends, the normality assumption doesn't seem very satisfied.

(D) The errors have constant variance (homoscedasticity)

decently satisfied. it's not perfect since there is a little bit of a variation between the spread on the residuals vs fitted plot. But since it doesn't look too bad and the Breusch-pagan p-value is above a 0.05 significance level we don't have enough convincing evidence that the error variance changes systematically

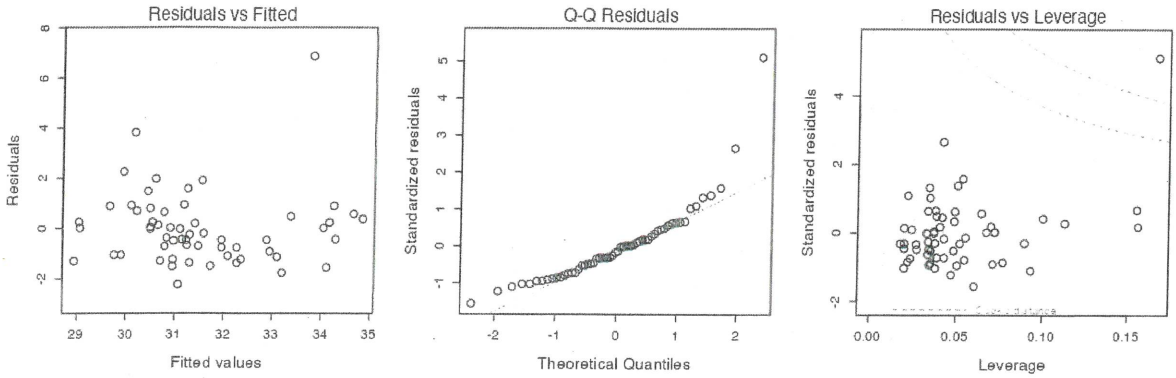
(E) There are no unduly influential points

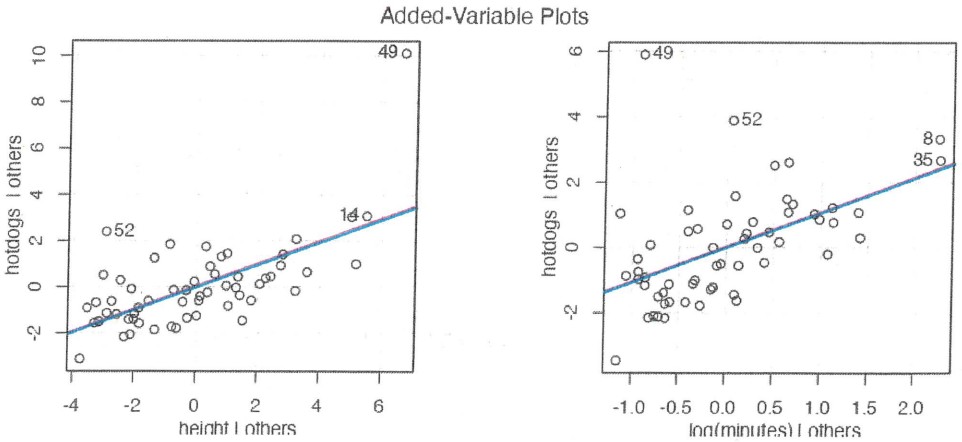
this looks ok. looking at the leverage vs. standardized residual plot, we see that we have a point w/ a high residual and a couple with high leverage, which have the potential to be influential, we don't have any points in the right hand corners which would show us influential points because influence (or cook's distance) depends on both a high residual and high leverage.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

not necessarily. just because model 1 has a better result from the breusch-pagan test that doesn't automatically make it a better model. it might satisfy the equal variance assumption slightly better but it still doesn't have a very good equal variance and model 2 improved some of the other assumptions that were violated in model 1. It's not an automatic passfail comparison based on one test, and we have to look at several other things to see what might be better overall. w/ unequal variance, the coefficient estimates are still unbiased when the mean model is right whereas in model one, the estimates aren't very reliable because linearity is violated. So model 1 is not automatically superior.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

advantages of model 2 include the fact that it made the linearity between log(minutes) and hot dogs satisfied, after accounting for height. so that estimated coefficients will be less biased and the interpretation will make more sense.

disadvantages include the fact that the residuals vs fitted plot still show a slight pattern so it might be better but not entirely linear. it also made the variances less equal based on the breusch pagan test. it didn't solve normality and now we have an influential point in the top right corner of the leverage vs. standardized residual plot

Question 16: State the estimated value of β_2 and provide an interpretation in context

$\hat{\beta}_2 = 1.06702$

Level-log interpretation

calculation

$\hat{\beta}\log(1+c/100)$ or $\approx \hat{\beta}/100$

$1.06702\log(1.01)$

$= .01061702$

Holding the height of a person constant, a 1% increase in a persons minutes per week they spent training is associated with an estimate 0.01061702 increase in mean (Average) Amount of Hot Dogs consumed in 10 minutes, on average.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\widehat{hotdogs} = .48834(height) + 1.06702\log(min)$

$= .48834(68) + 1.06702(4.369)$

$= 33.20712 + 4.66181038$

$= 37.86893038$

the predicted amount of hotdogs he will be able to eat is roughly 37.87.

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

$n-p-1$

$n=58 \quad p=2$

$58-3=55$

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1+x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$37.87 \pm t^* SE_{pred}$

$37.87 \pm 2.004(1.485)$

37.87 ± 2.97594

$(34.894, 40.846)$

we 95% confident that the true amount of hotdogs he'll be able to eat, based on his height being 68 inches and for training 79 minutes a week, is between 34.894 AND 40.846.

.16559

.10599

Part III - Application

Questions 20-24 concern the following dataset:

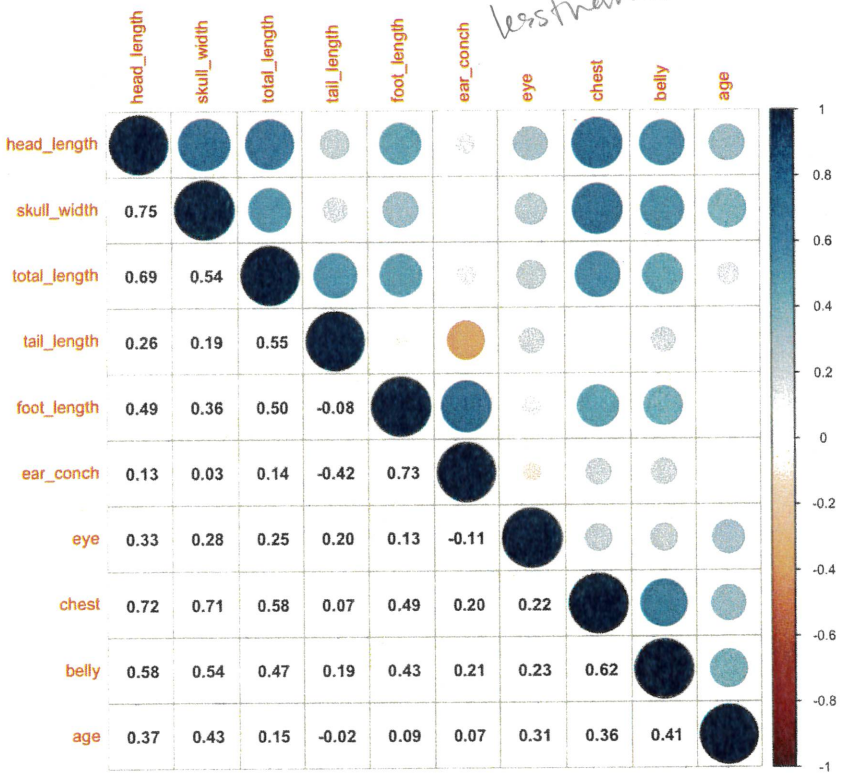
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
foot_length   -0.21499    0.10733  -2.003   0.0507 .
ear_conch      0.15730    0.10231   1.538   0.1306
eye            0.43736    0.22549   1.940   0.0582 .
chest          0.01883    0.18154   0.104   0.9178
belly          0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$\hat{\beta}_{\text{skull width}} = .16559$ $SE_{\hat{\beta}_{sw}} = .10599$

$0.16559 \pm t^* SE_{\hat{\beta}_{sw}}$

$0.16559 \pm (2.01)(.10599)$ $0.16559 \pm .2130399$

0.16559 ± 0.2130399
OR
$(-0.0474499, 0.3786299)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

holding a possum's head length, skull width, total length, tail length, foot length, ear conch circumfrance, right eye width, and chest girth constant, we are 95% confident that every cm of belly girth is associated with a change in age from a decrease of 0.023 years to an increase of 0.443 years on average

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

multicollinearity. since the results show us that at least one of the coefficients does not equal 0 and shows that the model has some explanatory value based on the F-test, but we don't have convincing evidence that a specific coefficient is actually not 0, when holding our predictors constant, the explanatory power likely comes from the relationship between the predictors, or multicollinearity

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

best subsets runs through all possible candidate models for the set of predictors we have, or fits a model with each combination of predictors we could make a model from and calculates AIC from each one using the formula $AIC = n\log(SSR/n) + 2k$ WHICH IS ASYMPTOTICALLY RELATED TO LEAVE ONE OUT CROSS VALIDATION WHICH MEANS AS $n \to \infty$, AIC IS GOING TO HAVE RESULTS CLOSER AND CLOSER TO WHAT WE GET FROM CROSS VALIDATION, meaning it's a good metric to use when we care about prediction. The best subsets algorithm chooses the model with the lowest AIC metric as the best model.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

a smaller one chosen by BIC could have a higher adj r^2

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM